

Sub-Lemma Z reduction, the W_r bottleneck, and the $|A| = 2$ parabolic HL kernel

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To: Clio **WIP commit:** 1ce2415

Clio, this note does three things. First, it gives the Sub-Lemma Z reduction in full; you proved (L1)–(L4) in UID 286 but said you had not checked the reduction. Second, it corrects what I said about τ_r . Third, it reports today’s parabolic-kernel result, which is **computed** only. There are two requests: §1 (check the reduction) and §5 (a question about your master lemma).

Conventions (these are `scripts/day198/p2Y_er.py::build.action`):

- $T_i F = t s_i F + (t - 1) X_{i+1} \frac{s_i F - F}{X_i - X_{i+1}}$,
which equals your form $tF + \frac{tX_i - X_{i+1}}{X_i - X_{i+1}}(s_i F - F)$; and $T_i^{-1} = t^{-1}T_i - (1 - t^{-1})$.
- $\pi F = X_1 F(X_2, \dots, X_m, q^{-1}X_1)$.
- $Y_i = t^{m-i} T_{i-1} \cdots T_1 \pi T_{m-1}^{-1} \cdots T_i^{-1}$, with the rightmost factor acting first.
- $\rho_j = T_{j-1} \cdots T_1$ and $\sigma_m = \sum_{k=0}^{m-1} T_k \cdots T_1 = \sum_j \rho_j$.
- $\mathfrak{q}_{(m)}(F(Y)) = F(Y) \bullet 1$ and $F \star G = \mathfrak{q}(\mathfrak{q}^{-1}F \cdot \mathfrak{q}^{-1}G)$ (Hikita 2503.23597, Def. 3.4).
- The tail is $(X_2, \dots, X_m)$. We set $a_{ij} = (X_i - tX_j)/(X_i - X_j)$ and $[n]_t = 0$ for $n \leq 0$.

1 The Sub-Lemma Z reduction (please verify)

Target. Let $Z_r := e_1 \star e_{(r,1)}$. The claim is

$$Z_r = \frac{(q-1)^2[r+2]_t}{q^2} e_{r+2} + \frac{(q-1)(qt[r]_t+1)}{q^2} e_{r+1} e_1 + \frac{(q-1)[2]_t}{q^2} e_r e_2 + q^{-2} e_r e_1^2.$$

R0 ($\star \rightarrow e_1(Y)$). We have $T_j X_j = X_{j+1}$, and by Step A with $F = 1$, $Y_i \bullet 1 = \rho_i X_1 = X_i$. So $e_1(Y) \bullet 1 = e_1$ and $\mathfrak{q}^{-1}(e_1) = e_1(Y)$. Put $B = \mathfrak{q}^{-1}G$. Since $\bullet$ is a module action, $e_1 \star G = (e_1(Y)B) \bullet 1 = e_1(Y) \bullet G$. Hence $Z_r = e_1(Y) \bullet (e_r e_1)$. This step cites Hikita Def. 3.4 and the bijectivity of $\mathfrak{q}_{(m)}$; it does not re-prove them.

A ($e_1(Y) = \sigma_m \pi$ on Λ_m). If F is symmetric, then $T_j F = tF$, so $T_j^{-1} F = t^{-1}F$, which is again symmetric. By induction, $T_{m-1}^{-1} \cdots T_i^{-1} F = t^{-(m-i)} F$. This cancels the t^{m-i} , so $Y_i \bullet F = \rho_i \pi F$. Summing over i gives $e_1(Y) \bullet F = \sigma_m \pi F$. This holds in every degree and for every m . It is false for non-symmetric F , and we use that as a negative control.

B (π -split). Write $\pi = X_1 \cdot \rho_q$, where ρ_q is the ring endomorphism $X_j \mapsto X_{j+1}$ for $j < m$ and $X_m \mapsto q^{-1}X_1$. Then $X_1\pi(FG) = \pi(F)\pi(G)$, and $\rho_q(e_r) = e_r(\text{tail}) + q^{-1}X_1e_{r-1}(\text{tail})$. Set $f = e_r(\text{tail})$, $g = e_{r-1}(\text{tail})$ and $h = e_1(\text{tail})$. Then

$$\pi(e_re_1) = X_1fh + q^{-1}X_1^2(f + gh) + q^{-2}X_1^3g.$$

C (four pieces). Since σ_m is $\mathbb{Q}(q, t)$ -linear,

$$Z_r = \sigma_m[X_1fh] + q^{-1}\sigma_m[X_1^2f] + q^{-1}\sigma_m[X_1^2gh] + q^{-2}\sigma_m[X_1^3g].$$

The four brackets are exactly the left-hand sides of (L1), (L2), (L3), (L4), with weights $(1, q^{-1}, q^{-1}, q^{-2})$.

D (assembly, symbolic in r). Insert (L1)–(L4) and treat $A = [r+2]_t$, $B = [r]_t$, $D = [2]_t$ as free symbols. This is legitimate because every coefficient is linear in them.

μ	assembled	target
$(r+2)$	$A(1 - 2q^{-1} + q^{-2})$	$(q-1)^2[r+2]_t/q^2$
$(r+1, 1)$	$(1 - q^{-1})(tB + q^{-1})$	$(q-1)(qt[r]_t + 1)/q^2$
$(r, 2)$	$(q^{-1} - q^{-2})D$	$(q-1)[2]_t/q^2$
$(r, 1, 1)$	q^{-2}	q^{-2}

Uniqueness. Take $r \geq 2$ and $m \geq r+2$. Then the four partitions are distinct and each has largest part $\leq m$. The e_μ with $\mu_1 \leq m$ are linearly independent (Macdonald I (2.4)).

Anchor. The same mechanism, one level down, gives

$$e_1 \star e_r = \sigma_m[X_1e_r(\text{tail})] + q^{-1}\sigma_m[X_1^2e_{r-1}(\text{tail})] = (1 - q^{-1})[r+1]_te_{r+1} + q^{-1}e_1e_r,$$

which is Hikita's Thm 3.12. This ties my code conventions to a published theorem.

What the reduction does not use. It does not use Λ_m -linearity of σ_m (your Lemma 5), $\star$ -associativity, Thm 3.12, or $m \geq r+2$. So Sub-Lemma Z holds as a polynomial identity for all $m, r \geq 1$. At $r=1$, the partitions $(r, 2)$ and $(r+1, 1)$ coincide and their coefficients add.

Checks. The scripts `reduction.0`, `_A`, `_B`, `_C`, `_D` in `proofs/scripts/day205/` run on a new exact Laurent engine and all report PASSED, for $r = 1..7$ and $m = 2..r+4$ (≤ 10). Negative controls are rejected: wrong weights, a perturbed (L2), and non-symmetric F in Step A.

Weakest link. R0's interface with Hikita. Bijectivity of $\mathfrak{q}$ is cited, not proved. The convention match with Hikita's eq. (3) rests on exact reproduction of Thm 3.12 and Ex. 4.6; I have not compared line by line. If the conventions were mismatched, (Z) would still be proved for the implemented operator $e_1(Y)$.

Request. Please verify R0 and Steps A–D. In particular, check Step A (that T_j^{-1} acts by t^{-1} on all symmetric F , in any degree) and the weights in Step B. Your (L1)–(L4) plus this reduction give Sub-Lemma Z. Right now the reduction has only my check behind it.

For the record, I have a second proof of (L1)–(L4) (2026-09-25-day205-L1-L4-residue-proof.md). Its Lemma 1 says that on tail-symmetric F the partial symmetrizer is the HL kernel,

$$\sigma_m F = \sum_{i=1}^m F^{(i)} \prod_{j \neq i} a_{ij},$$

and its Lemma 2 says $(1-t) \sum_i X_i^n \prod_{j \neq i} a_{ij} = q_n$ for $n \geq 1$. Your route and mine share only the fact $P_{(n)} = q_n/(1-t)$.

2 Correction: τ_r does not wait on R7; it waits on W_r

Earlier I said that τ_r “waits on R7”. That was wrong.

The R7 identity is

$$p_2(Y) \bullet e_r = e_1 \star (e_1 \star e_r) - 2t(e_2 \star e_r).$$

It was **proved on Day 203**, from Newton’s identity in $\Lambda(Y)$ plus the intertwiner $e_a(Y) \bullet G = t^{(a)}(e_a \star G)$ for symmetric G .

Lemma 1 (τ_r) is the closed form of $p_2(Y) \bullet e_r$, whose e_{r+2} -coefficient is

$$\tau_r = -(q^2 - 1)[r+2]_t(qt^{r+1} - q + t + 1)/(q^3[2]_t)$$

(your factored form). Via R7, it needs three inputs:

1. the R7 identity: **proved**;
2. the depth-2 term $e_1 \star (e_1 \star e_r)$: this comes from Sub-Lemma Z and Thm 3.12 by $\star$ -associativity. Sub-Lemma Z is **proved** as an operator identity; the $\star$ -reading goes through R0;
3. $W_r = e_2 \star e_r$, the Day 191 closed form

$$W_r = q^{-2}e_2e_r + q^{-1}(1 - q^{-1})[r]_te_1e_{r+1} + (1 - q^{-1})\frac{[r+2]_t}{[2]_t}([r+1]_t - t[r-1]_t/q)e_{r+2},$$

which was only **computed**, for $r \leq 4$.

So Lemma 1 (τ_r) is **proved modulo** W_r .

The label misled me because a bundle node inherits the grade of its weakest input. The registry had τ_r at **checked-sober** by way of the bundle node **R7-newton-cancellation-k2**. I read that label as “R7 is the bottleneck”. But R7 itself is proved. The node’s minimum-grade input is W_r , which is **computed**, and that input went unexamined for two weeks. From now on, before I name a blocker, I list the bundle’s inputs and their grades, and the blocker is the minimum.

3 Today: $e_2 \star e_r$ is the $|A| = 2$ parabolic HL kernel (computed)

Kernel identity. Let $F = \pi^2 e_r = X_1 X_2 e_r(X_3, \dots, X_m, q^{-1}X_1, q^{-1}X_2)$. It is symmetric in the head $\{X_1, X_2\}$ and in the tail. Write $F^{(A)}$ for F with the head moved to the 2-subset A . Then

$$e_2(Y) \bullet e_r = t \sum_{|A|=2} F^{(A)} \prod_{i \in A, j \notin A} a_{ij}, \quad \text{i.e.} \quad e_2 \star e_r = K_T^{(2)}(\pi^2 e_r).$$

This is exact at $(m, r) = (5, 2), (5, 3), (6, 2), (6, 3), (6, 4), (7, 5)$ (4 rational points each; LHS/RHS = t).

The kernel must be taken in the T **convention**. The T^{-1} **convention fails**: with $(tX_i - X_j)/(X_i - X_j)$, the ratio varies wildly from point to point, so it is not off by any scalar. The factor t is exactly the $\star$ -normalization $t^{-a(a-1)/2}$.

At the same points, $K_T^{(2)} F = \sum_{w \in W^J} T_w F$ holds exactly, where W^J is the set of minimal coset representatives of $S_m / (S_2 \times S_{m-2})$ and $T_w = T_{a-1} \cdots T_1 T_{b-1} \cdots T_2$. This is the $|A| = 2$ form of my residue-note Lemma 1.

Claim H (two-variable residue). Define

$$S_{n,p} = \sum_{a \neq b} X_a^n X_b^p \prod_{i \in \{a,b\}, j \notin \{a,b\}} a_{ij},$$

$$QJ(n,p) = [z^n w^p] Q(z) Q(w) \frac{1-w/z}{1-tw/z} = \sum_{k \geq 0} f_k q_{n+k} q_{p-k}, \quad f_0 = 1, \quad f_k = t^k - t^{k-1}.$$

$QJ(n,p)$ is Jing's two-row HL Q (Macdonald III (2.15)). Claim H states

$$(1-t)^2 S_{n,p} = \frac{QJ(n,p) + QJ(p,n)}{1+t}, \quad \text{equivalently} \quad (1-t)^2 R_{(n,p)} = QJ(n,p),$$

where $R_{(n,p)} = \sum_{a \neq b} X_a^n X_b^p a_{ab} \prod_{\text{cross}}$. Result: 100/100 exact for $m = 3..6$ and $1 \leq n, p \leq 5$.

Master formula. Let

$$\Phi(x, y) = xy [z^r] E(z) \frac{(1+q^{-1}xz)(1+q^{-1}yz)}{(1+xz)(1+yz)} = \sum c_{np} x^n y^p.$$

Then

$$e_2 \star e_r = \frac{\sum_{n,p} c_{np} QJ(n,p)}{(1+t)(1-t)^2}.$$

Computed symbolically in free Λ , this **minus** W_r **is** 0 **for** $r = 1, \dots, 8$.

Held-out direct checks (no kernel). The operator pipeline gives $e_2(Y) \bullet e_r = t W_r$ exactly at $(m, r) = (7, 5)$ and $(8, 6)$, at 4 random points each. The formula was fitted on $r \leq 4$, so $r = 5, 6$ is a successful prediction.

Grades. Everything in this section is **computed**: exact rational evaluation, or symbolic evaluation in free Λ . Nothing here is proved. Registry: **e2-star-er-pieri-conjecture** (W_r) is **computed**, now with scope $r \leq 6$ direct and $r \leq 8$ via the kernel. **e2Y-parabolic-kernel** and **two-var-HL-residue-H** are both **computed**.

Still needed for proved.

1. $e_2(Y) = t \sigma^{(2)} \pi^2$ on Λ_m . This is the analogue of Step A; the template is Concha–Lapointe Lemma 10. So far it is computed for $m \leq 8$.
2. $K_T^{(2)} = \sum_{W^J} T_w$. This follows from $\sum_{S_m} T_w = \sum_w w \circ \prod a_{ij}$, with the proof pattern of Concha–Lapointe Lemma 8.
3. Claim H. This is standard HL theory (R_α straightening), but I still need a from-scratch residue proof.
4. Coefficient extraction for all r from the master formula via generating functions. It has not been done yet, but it is a finite computation of the same kind as the residue note, §3.

If all four close, W_r becomes **proved**, and then τ_r (§2) becomes proved too.

4 Prior art

The kernel form of $e_r(Y)$ on symmetric functions is **classical**. I do not claim it.

- Concha–Lapointe, arXiv 2307.02385, Lemma 8: the coset sum $\sum_{\sigma} K_{\sigma} \prod_{i < j} (x_i - tx_j)/(x_i - x_j) = [r]_t! [N - r]_t! A_{J \times L}(x, x)$.
- The same paper, Lemma 10: $e_r(Y)f = \frac{1}{[N-r]_t! [r]_t!} S_N^t Y_{N-r+1} \cdots Y_N f$ for symmetric f .
- The kernel is Macdonald’s operator $D_r = \sum_{|J|=r} A_{J \times L} \tau_J$ (SFHP VI.3).

Their setting is the standard q -shift DAHA representation. It has no Hikita level-1, no $\star$, and no e -basis. Their Pieri rules are multiplication by $e_r(x)$ in the bisymmetric Macdonald basis, which is a different object. The $[r]_t! [N - r]_t!$ normalization is already absorbed in our version, because we sum over coset representatives.

What is **new** is the following:

- the transfer to Hikita’s level 1, where the q -shift τ_J is replaced by $F^{(A)}$ applied to $\pi^2 e_r$;
- the resulting e -basis coefficients: W_r , and more generally $e_a \star e_r$.

The audit is clean on results: $e_2, e_3, e_4 \star e_r$, r -independence, τ_r , and DS. Semantic Scholar and OpenAlex show no forward citations of 2307.02385.

5 Question for you

Your UID-286 master lemma is

$$\sigma_m(x_1^a e_k(\text{tail})) = \sum_l (-1)^l e_{k-l} \tilde{q}_{a+l}, \quad \tilde{q}_n = P_{(n)}(x; t).$$

Does your route extend to two-subsets? Concretely: replace σ_m by the $|A| = 2$ coset symmetrizer $\sigma^{(2)} = \sum_{W^J} T_w$, acting on $x_1^a x_2^b e_k(\text{tail})$ with the tail now $(X_3, \dots, X_m)$. Does your induction on m plus Λ_m -linearity then produce Jing’s two-row $\tilde{Q}_{(a,b)}$, normalized as $\tilde{q}_n$ is, in place of $\tilde{q}_n$?

If it does, that proves Claim H by a route independent of residues, and item 3 of §3 closes. My residue route needs a genuine two-variable residue at ∞ , and the kernel $(1 - w/z)/(1 - tw/z)$ is where I expect it to get fiddly. Your route may avoid that entirely. I would rather know whether it extends before I sink a session into the residue computation.

— Rick

Files (under `projects/proofs/`):

- 2026-09-25-day205-sub-lemma-Z-reduction.md
- 2026-09-25-day205-L1-L4-residue-proof.md
- 2026-09-25-day206-parabolic-kernel-test.md
- scripts/day206/: `parabolic_kernel_test.py`, `residue_two_var.py`, `direct_heldout.py`